

assignment more trig and vectors fun - make up oct 3rd



⚠ This is a preview of the draft version of the quiz

Started: Apr 29 at 7:30pm

Quiz Instructions

There is no tutorial. No class the 3rd.

remember: you have tutorials about vectors in the dropbox. Go over textbook

Remember the head of the vector points to its (x,y) coordinates when the tail is attached to (0,0)



Question 1 5 pts



ha is 60m. Using a protactor you are measuring the angle as shown. What is the height of the building?
round to the nearest ten. The picture is not on scale



about 60m



about 70m



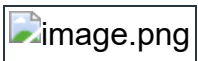
about 100m



about 80m



Question 2 5 pts



What is the angle theta? round to nearest 10. The picture is not on scale



10 degrees

☐

30 degrees

☐

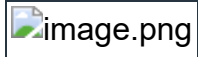
20 degrees

☐

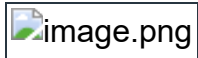
5 degrees



Question 3 5 pts



I am helping you below.



What is the standard notion for the vector velocity relative to shore? You can use the polar graph (header) and trace your vector.

☐

4.5 m/s @ 27

☐

2m/s @ 27

☐

4.5 m/s @ 63

☐

4m/s @ 63



Question 4 5 pts



that is what is the velocity of the boat relative to the shore. You can draw the vector in a polar graph. You have its coordinates. watch for negative sign. Take the angle from +xaxis. CCW is positive but CW is negative

☐

6.8 m/s @ -51

☐

6.8m/s@ - 39

☐

6.8m/s @ 51

☐

6.8m/s @ 39



Question 5 5 pts

how long for the boat to cross the river? The river is 1800 m wide.

hint: use only one of the 2 components

☐

7.5 minutes

☐

6.7 minutes

☐

15 minutes

☐

can not be determined



Question 6 5 pts



image.png

round to the nearest ten. picture is not on scale

☐

510 feet

☐

110ft

☐

20ft

☐

310 ft



Question 7 5 pts



image.png

The component of vector $\mathbf{g}$?

You can attach the tail to the origin of a (X,Y) coordinate system. The head shows the coordinate. Or move from tail to head. One square is 1 unit

☐

(-2,-4)

☐

(-2, 4)

☐

(2,4)

☐

(2,-4)



Question 8 5 pts

building on previous question. You can use a polar graph for this question. See header.

What is the standard notion for vector g ? (**angle is measured CCW from positive x-axis**)

round angle to nearest 10

☐

$g=2$ units @ 60 degrees

☐

$g=4.5$ units @ 30 degrees

☐

$g=4.5$ units @ 120 degrees

☐

$g=4$ units @ 150 degrees



Question 9 5 pts

building on previous question.

Consider the vector sum $S = g + a$

What is the coordinates of the vector and its angle relative to the positive x-axis (around to nearest 1)

(review adding vectors algebraically)

☐

$S(2,2)$ and 45 degrees

☐

$S(2,6)$ and 72 degrees

☐

$S(2,6)$ and 16 degrees

☐

$S(6,6)$ and 45 degrees



Question 10 5 pts

Consider the vector g and the vector c .

What is the coordinates of the vector sum S and its angle relative to the + x-axis?

use a polar graph in header for help.

☐

S(0,3) and 90 degrees

☐

S(-4,3) and 127 degrees

☐

S(-4,3) and 143 degrees

☐

S(-4,3) and 37

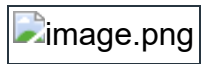
☐

S(4,3) and 37 degrees



Question 11 5 pts

here are 3 vectors:



What is the coordinates of the vector sum and its angle with the +x-axis ? (use a graph if you want)

You can do it using the components. Remember. The head of a vector shows its component (tail attached to origin). You add components.

☐

S(-25,10) and 158 degrees

☐

S(-15,10) and 34 degrees

☐

S(-15,10) and 124 degrees

☐

S(-15,10) and 146 degrees



Question 12 5 pts



What is the coordinate of the vector sum?

☐

S(15,0)

☐

S(-15,10)

☐

S(25,20)

☐

S(-15,0)



Question 13 5 pts



So diagram 1

☐

a

☐

d

☐

b

☐

c



Question 14 5 pts

Building on previous question. what about diagram 3

☐

b

☐

c

☐

d

☐

a



Question 15 5 pts

Building on previous question.

Which diagram shows: $A+B+C=0$

☐

diagram 3

☐

disgram 2

☐

diagram 4

☐

disgram 1



Question 16 5 pts



Which vector represents $B - A$? (about)



You can use $B - A = B + (-A)$ or you can go from A to B if they are attached by their tails.



c



d



b



a



Question 17 5 pts



!!! $-(-)$ is +

First subtract the coordinates then find the magnitude (length of arrow in the (x,y) system)



7



8



11



9



10



Question 18 5 pts



The scalar product is also called the dot product. To find the dot product :

$$x_A x_B + y_A y_B + z_A z_B$$

for example $x_A=6$, $x_B = 5$ so $x_A x_B=30$



60



0



12



48



Question 19 5 pts

Building on previous question what is the angle between A and B?

hint: dot product = $|A| |B| \cos(\text{angle between A and B})$

magnitude of a vector is found using pythagorean theorem. for example $|A| = \sqrt{x_A^2 + y_A^2 + z_A^2}$
no minus inside.

the dot product was found in the previous question)



about 10 degrees



about 50 degrees



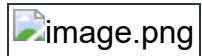
about 20 degrees



about 80 degrees



Question 20 5 pts



B



D



C



A

Not saved

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